Frontier



Clean Energy Conversions Laboratory

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Clean Energy Conversions Laboratory, of the University of Pennsylvania Department of Chemical and Biomolecular Engineering and the Kleinman Center for Energy Policy.

Organization location (we welcome applicants from anywhere in the world)

Philadelphia, Pennsylvania, United States

Name(s) of primary point(s) of contact for this application

Abby Lunstrum, Jennifer Wilcox

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

University research lab affiliated with the department of chemical engineering and climate policy center

1. Public summary of proposed project[1] to Frontier

a. **1A Project title** <20 words

Co-extraction of hydroxides and critical minerals from industrial wastes for cost-efficient CDR

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. 1B Project summary and field impact:

- Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Our core research goal is to lower the cost of alkalinity-based CDR by engineering systems for the co-recovery of mixed hydroxides and critical metals from industrial wastes.

Surficial mineralization is potentially a highly scalable CDR method given large volumes of alkaline feedstock from mine wastes and industrial byproducts (Power et. al., 2013; Renforth, et. al., 2019). However – as identified in the RFP – most raw feedstocks are slow to mineralize on their own, and activation is energy-intensive and expensive. This kinetic barrier results in a fundamental tradeoff between capture rates and cost. While novel approaches like microwave pretreatment can marginally increase reaction rates, there are no obvious pathways available to speed up mineralization rates sufficiently for cost-effective, large-scale CDR. We propose an alternative approach: to **reduce costs** by combining the **recovery of valuable critical metals (CM) alongside reactive alkaline minerals** (mixed hydroxides) from industrial waste products. By valorizing CM and utilizing legacy waste stockpiles, costs incurred by the extraction of purified hydroxides are minimized, and in some cases potentially negligible. This approach reframes alkalinity-based CDR from a cost-limited pathway to a potentially revenue-supported system, contingent on feedstock composition and recovery efficiency.

Our proposed work involves four steps. First, we will use well-established hydrometallurgy techniques to extract CM, followed by purified, reactive alkaline minerals (e.g., $Ca(OH)_2$ or $Mg(OH)_2$) from industrial wastes. We will initially test coal fly ash, which is abundantly available in legacy stockpiles, as well as asbestos waste. For both feedstocks, we will test energetically mild pre-treatment approaches (e.g., microwave) to facilitate faster reaction kinetics. We will measure process inputs, recovery efficiencies, and product purity to inform CDR cost analyses in later steps. Second, we will test the produced hydroxides as feedstock for **accelerated surficial mineralization**, using CO_2 concentrations in ambient air. This work will involve measuring mineralization kinetics in controlled environments with e.g. varying temperature and humidity. In addition to surficial mineralization, we will also assess the hydroxide product's suitability for ocean alkalinity enhancement (OAE) applications. If the hydroxides produced in Step One are of acceptable purity (e.g., low trace metal concentration), it can potentially be used in distributed applications like OAE. While surficial mineralization is advantageous in terms of relatively straightforward MRV, OAE applications can sequester nearly twice the amount of atmospheric CO_2 per mass of material. This **maximized efficiency effectively further lowers the cost of CDR**. This work will include measuring dissolution kinetics and stability thresholds (i.e., the maximum rate at which it can be added to seawater before inducing secondary precipitation). Third, we will conduct a comprehensive lifecycle assessment (LCA) and technoeconomic analysis (TEA) – considering both CapEx and OpEx – to assess the net cost of CDR via both methods. These analyses will consider the co-recovery of CM, as well as environmental remediation benefits, as options to offset the cost of CDR. They will also include a comprehensive sensitivity analysis to understand the conditions under which CDR costs are lowest (considering, for example, feedstock chemistry, transport distances, energy sourcing, etc.). Fourth, we will conduct an inventory and geographic analysis of relevant waste stockpiles, including both coal fly ash and asbestos, to assess scalability and identify the most cost-efficient locations to deploy the technology. This analysis will also consider storage resources (e.g., mines or oceans).

Our approach aligns with several key points in the RFP: The integration of **advanced electromagnetic**

feedstock preparation techniques via microwave pretreatment can enhance CM, REE, and alkalinity recovery. Rather than employ microwaves to enhance carbonation of coal ash and asbestos feedstocks, we employ this pretreatment technique to increase extraction efficiency at the leaching phase, allowing for lower reagent costs and embodied emissions. Our approach represents a **disruptive sourcing innovation** by taking advantage of the US's vast deposits of industrial waste. Coal ash ponds and landfills represent a feedstock of more than 1 billion tons landfilled between 1991 and 2016 (DOE 2022). This can be supplemented by fresh coal fly ash produced each year: in 2022, the US produced more than 28 million short tons of coal fly ash (and as much as 50 million tons in 2014) (ACAA 2025). It is estimated more than 2 billion tonnes of chrysotile asbestos mine waste remains unremediated across the US and Canada (Sandalow et al., 2021).

Benefits to CDR field:

This project introduces a cost-reduction pathway for alkalinity-based carbon dioxide removal by coupling carbon removal with revenue-generating critical mineral recovery. By shifting the economic model from a pure cost center to a co-product system, this work has the potential to significantly reduce net CDR costs under favorable feedstock conditions.

In addition, the project generates foundational data on the performance of mixed Ca–Mg hydroxides, which is an underexplored but highly relevant class of materials for both surficial mineralization and ocean alkalinity enhancement.

Finally, by integrating experimental results with technoeconomic, lifecycle, and geospatial analysis, this work produces decision-relevant insights on where and under what conditions this pathway is viable at scale, directly addressing key uncertainties in deployment.

References:

ACAA. (2025). *Beneficial Use of Coal Combustion Products: An American Recycling Success Story*. https://acaa-usa.org/wp-content/uploads/2022/12/22-ACAA-Brochure_FINAL.pdf

DOE. (2022). *Recovery of Rare Earth Elements and Critical Materials from Coal and Coal Byproducts*. US Department of Energy. https://www.energy.gov/sites/default/files/2022-05/Report%20to%20Congress%20on%20Recovery%20of%20Rare%20Earth%20Elements%20and%20Critical%20Minerals%20from%20Coal%20and%20Coal%20By-Products.pdf

Power, I. M., Harrison, A. L., Dipple, G. M., Wilson, S., Kelemen, P. B., Hitch, M., & Southam, G. (2013). Carbon Mineralization: From Natural Analogues to Engineered Systems. *Reviews in Mineralogy and Geochemistry*, *77*(1), 305–360. https://doi.org/10.2138/rmg.2013.77.9

Renforth, P. (2019). The negative emission potential of alkaline materials. *Nature Communications*, *10*. https://doi.org/10.1038/s41467-019-09475-5

Sandalow, D., Aines, R., Friedmann, J., Kelemen, P., McCormick, C., Power, I., Schmidt, B., & Wilson, S. (2021). *Carbon Mineralization Roadmap*. ICEF Innovation Roadmap Project. https://www.icef.go.jp/wp-content/uploads/2024/02/icef2021_roadmap.pdf

c. 1C Project objectives:

Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

We seek to answer the following **research questions**:

- Can alkaline industrial wastes be efficiently processed to produce more reactive alkaline material – e.g., mixed $CaMg(OH)_2$ – suitable for CDR applications?
- How efficient are the mixed hydroxides at capturing CO_2 via passive mineral carbonation and via OAE?
- Does the co-recovery of mixed hydroxides and critical metals lower costs sufficiently to valorize CDR?
- How scalable is this process, considering the availability of industrial wastes such as coal ash and asbestos?
- What are the ideal locations to deploy this technology, considering feedstock availability, energy, and storage options.

We will answer the research questions with a methodology based both on **experimental** and **modeling** components, consisting of four key steps.

Step 1. The crux of our work is based on synthesized hydroxides from coal fly ash and asbestos waste. We produce this material through acid extraction of metal cations and sequential precipitation with sodium hydroxide (NaOH). Both acid and base reagents are regenerated in a circular, electrochemical process. This experimental procedure is illustrated in Figure 1 below. In addition to this baseline hydrometallurgical process, we will also test limited pre-treatment steps that can potentially facilitate faster extraction with minimal energy inputs. These steps include grinding, magnetic separation, and short-duration microwave pre-treatment. Microwave pretreatment has been found to enhance REE and Ca/Mg extraction efficiency from amorphous coal ashes with short bursts of microwave energy (Yakaboylu et al., 2019). Our preliminary results suggest short time scales of moderate power levels show best extraction enhancements, motivating our argument that microwave pretreatment is low energy, particularly when powered by low carbon electricity.

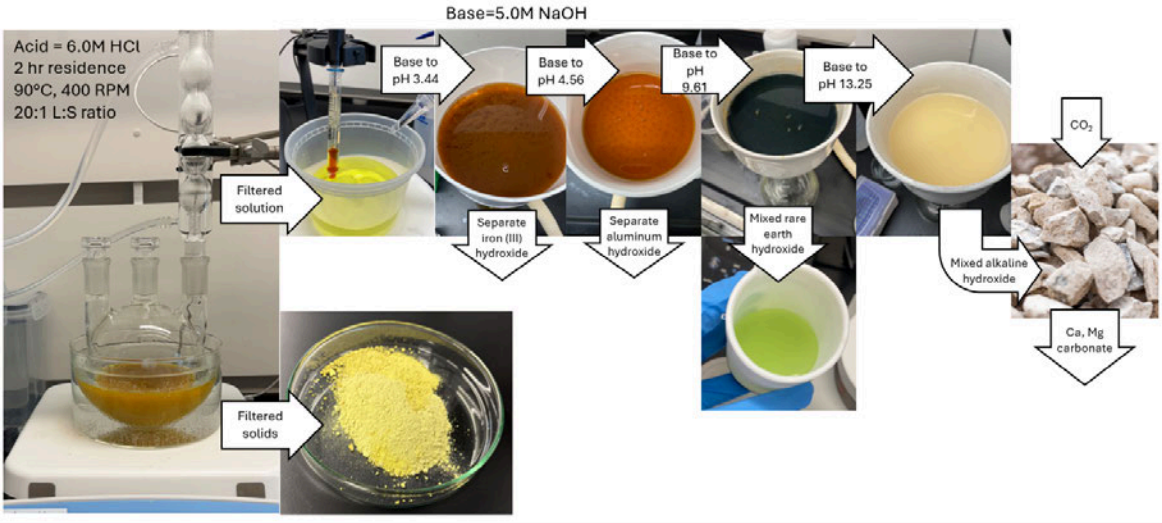


Figure 1. Experimental work flow to synthesize reactive alkalinity from coal ash and asbestos feedstocks

One product of acid extraction is a leaching liquor rich in metal cations. This concentration is dependent on both the elemental and mineralogical composition of the feedstock and the extraction conditions (temperature, residence time, solid to liquid ratio, acid concentration). The chemical makeup of the leaching liquid is shown in Figure 2. The residue of the extraction process is a ferro-allumino silicate solid. This waste product is safer for long-term storage than the original feedstock, as it is depleted of labile metals.

Element	Concentration	Element	Concentration
Na	0.64 g/L	Ge	0.16 ppm
Mg	2.38 g/L	La	1.63 ppm
Al	0.82 g/L	Ce	3.22 ppm
Si	0.08 g/L	Pr	0.39 ppm
K	0.12 g/L	Nd	1.90 ppm
Ca	15.85 g/L	Sm	0.30 ppm
Ti	0.83 g/L	Eu	0.07 ppm
Cr	0.01 g/L	Gd	0.28 ppm
Mn	0.03 g/L	Dy	0.25 ppm
Fe	3.39 g/L	Ho	0.05 ppm
Co	2.13 ppm	Er	0.14 ppm
Ni	4.44 ppm	Tm	0.02 ppm
Cu	13.75 ppm	Lu	0.02 ppm
Zn	8.79 ppm	U	0.22 ppm

Figure 2. Aqueous cation concentration in leaching liquor. Leaching conditions are for the Powder River Basin subbituminous ash combusted at the Scherer power plant in Georgia. Ash is contacted with acid in a 1:20 solid to liquid ratio.

Step 2. Synthesized hydroxides will be produced on the tens of grams **scale** and tested for CDR applications via a) passive carbonation in air, and b) dissolution in seawater. Passive carbonation

experiments will be modeled after Criado & Abanades, 2022; Dostie et al., 2025; McQueen et al., 2020; and Rausis et al., 2022. Experiments will be run at a range of humidity and temperature levels to assess carbonation rates and efficiency across natural environmental conditions. Complete carbonation of Ca and Mg hydroxides can take anywhere from hours to a year, depending on feedstock qualities and the design of the carbonation reactor. We will inform our experiments to test conditions to maximize carbonation rates. Both hydroxide feedstock and carbonated products will be characterized (e.g., metal concentrations, mineralogy, and adsorptive qualities, etc.) to inform our understanding of carbonation kinetics across feedstocks and environmental conditions, as well as guide storage considerations. Ca- and Mg-based hydroxides produce different carbonation products (e.g., calcite, hydromagnesite), with different storage and processing considerations, so assessing the storage potential of mixed products is critical. OAE experiments will be modeled after Varliero et al., 2024; and Moras et al., 2022 and 2024. These experiments will measure the dissolution kinetics of the synthesized hydroxides in seawater. Understanding how quickly the material dissolves is important for OAE deployment design, and ensuring that dissolution proceeds completely before particles sink out of the ocean's surface layer. The experiments will also test for the stability of alkalinity in longer-term (days to months) experiments. In high alkalinity conditions, following high rates of feedstock dosing, alkalinity may re-precipitate out of solution as CaCO3. The rate and likelihood of this instability varies with feedstocks, depending on particle surface properties that can support mineral nucleation. Understanding the thresholds for such loss to precipitation helps constrain deployment rates and informs scalability of using our synthesized hydroxides for OAE applications.

Step 3. The work will also include a comprehensive cradle-to-gate TEA and LCA model. This addition fills a major gap in scientific understanding, and our team is uniquely suited to address this research gap. The LCA and TEA models allow for variation of coal ash and asbestos feedstocks; energy and reagent costs and carbon intensities; transport distances; and market volatility of coproduct value. The LCA accounts for on-site energy consumption for running the process and the embodied energy of material inputs to unit construction (e.g., resource requirements and associated emissions of steel production to construct on-site ball mills). Accounting for on-site and embodied emissions allows for a more comprehensive accounting of greenhouse gas (GHG) emissions of the process scheme. In the TEA, we track costs of energy, reagent, water, and labor in the form of operating expenditure (OPEX) and capital investment (CAPEX). The model also incorporates value added through coproduct accounting and potential carbon credits for CO_2 stored via mineralization. Importantly, the models account for possible counterfactual feedstock uses. For example, some classes of coal fly ash can be used as a supplementary cementitious material (SCM), which achieves CO_2 avoidance by displacing ordinary Portland cement. Considering possible counterfactuals ensures our approach achieves positive additionality. The functional unit for the model is 1 tonne of ash or asbestos processed. The process-model core of the LCA and TEA is illustrated in Figure 3.

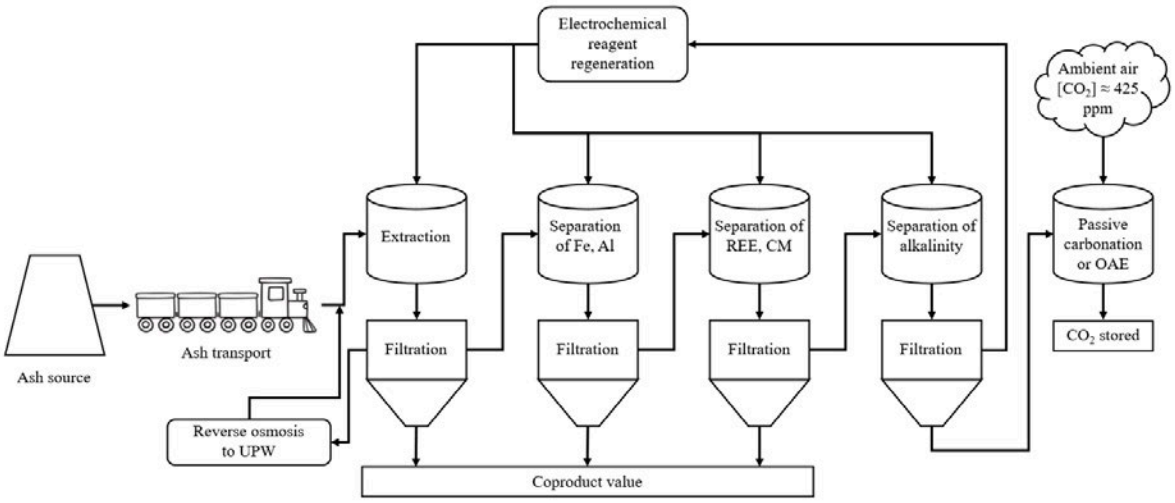


Figure 3. Techno economic and life cycle analysis schematic

Step 4. We will also build a national inventory of coal ash and asbestos stockpiles, and geospatially analyze this dataset considering cost-efficient locations for CDR technology deployment, low-carbon energy, and storage resources like mines and oceans. We have already begun this process for specific locations in Kentucky, Alabama, and Georgia (Figure 4A,B). Through the course of this work, we will expand nationwide and create tiers of feasible scalability: for coal ash, analyzing fresh, currently operating ponds, and closed ash ponds. In Figure 4C, we show an example dataset for closed ash ponds that have been proposed as harvesting sites.

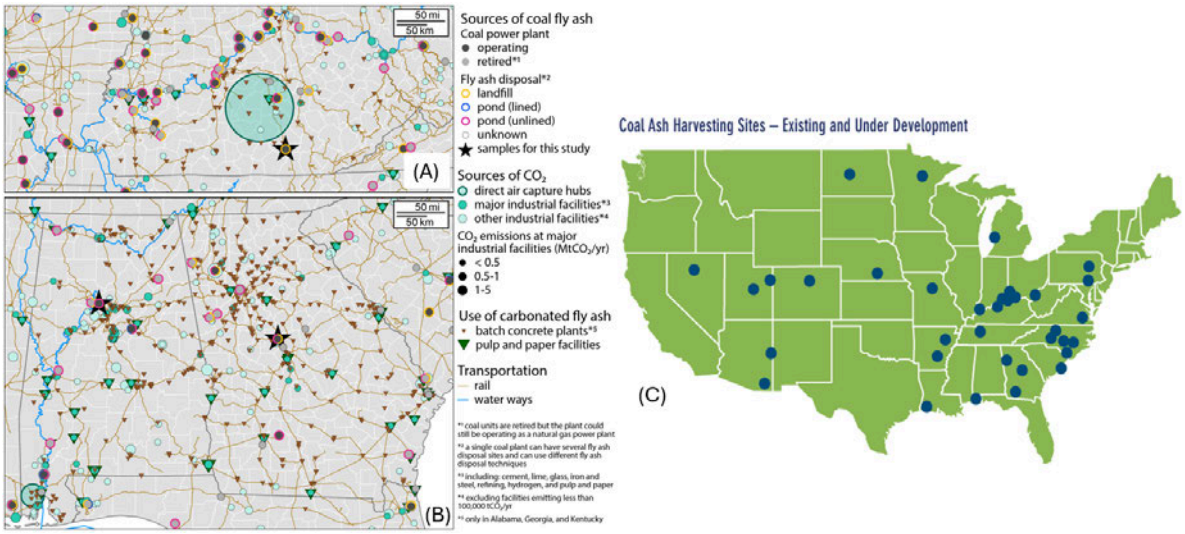


Figure 4. Maps of Kentucky (A), and Alabama and Georgia (B), showing the location of analyzed coal fly ash samples, along with other potential sources of coal fly ash and disposal sites, potential major sources of CO_2 including direct air capture hubs, and utilization opportunities for the carbonated coal fly ash in the construction and paper industries (data for A and B from 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, for C from ACAA, 2025).

In summary, we will accomplish the outlined **research objectives** through a coupled LCA and TEA, informed by experimental results, and integrated with disparate geographic data resources. This effort will result in the following tangible results:

1. Scientific progress on carbonation of mixed $CaMg(OH)_2$ synthesized from industrial waste products, with co-recovery of CMs
2. LCA and TEA models and results, including comparison to alternative use cases like SCM
3. Scalability and technical potential analysis illustrating hot spots for CDR citing based on production efficiency, current production rates and landfilled wastes, and proximity to storage.

Our team is primed to complete these research objectives because of the previous work conducted on passive air capture and OAE and our current expertise of LCA and TEA modeling of coal ash feedstocks including critical mineral and REE recovery, technical potential modeling of national scale coal ash mineralization, and geospatial analysis. The confluence of these skillsets makes us the **best in class** for this work. We are uniquely suited to bridge the research gap between basic science understanding of alkalinity and critical mineral extraction from industrial wastes and nationwide application scale of this integrated CDR pathway. The concept of coupling what would have otherwise been a prohibitively expensive alkalinity synthesis process with the co-recovery of energy critical and *valuable* critical metals and REE is already novel. The addition of our deep qualifications in passive air capture, OAE, and geospatial and scalability analysis creates a project that no other organization could accomplish.

References:

ACAA. (2025). *Beneficial Use of Coal Combustion Products: An American Recycling Success Story*. https://acaa-usa.org/wp-content/uploads/2022/12/22-ACAA-Brochure_FINAL.pdf

Criado, Y., & Abanades, J. C. (2022). Carbonation Rates of Dry Ca(OH)2 Mortars for CO2 Capture Applications at Ambient Temperatures. *Industrial and Engineering Chemistry Research*, *61*(40), 14804–14812. https://doi.org/10.1021/acs.iecr.2c01675

Dostie, L., Power, I. M., & Rausis, K. (2025). Carbon dioxide supply and scaling constraints on direct air capture using calcium oxide powder. *Journal of Cleaner Production*, *527*. https://doi-org.proxy.library.upenn.edu/10.1016/j.jclepro.2025.146635

McQueen, N., Kelemen, P., Dipple, G., Renforth, P., & Wilcox, J. (2020). Ambient weathering of magnesium oxide for CO2 removal from air. *Nature Communications*, *11*. https://doi.org/10.1038/s41467-020-16510-3

Moras, C. A., Bach, L. T., Cyronak, T., Joannes-Boyau, R., & Schulz, K. G. (2022). Ocean alkalinity enhancement – avoiding runaway CaCO3 precipitation during quick and hydrated lime dissolution. *Biogeosciences*, *19*(15), 3537–3557. https://doi.org/10.5194/bg-19-3537-2022

Moras, C., Cyronak, T., Bach, L., Joannes-Boyau, R., & Schulz, K. G. (2024). *Effects of grain size and seawater salinity on magnesium hydroxide dissolution and secondary calcium carbonate precipitation kinetics: Implications for ocean alkalinity enhancement*. *21*(14), 3463–3475. https://doi.org/10.5194/bg-21-3463-2024

Rausis, K., Stubbs, A. R., Power, I. M., & Paulo, C. (2022). Rates of atmospheric CO2 capture using

magnesium oxide powder. *International Journal of Greenhouse Gas Control*, *119*. https://doi-org.proxy.library.upenn.edu/10.1016/j.ijggc.2022.103701

Varliero, S., Buono, A., Carerini, S., Raos, G., & Macchi, P. (2024). Chemical Aspect of Ocean Liming for CO2 Removal: Dissolution Kinetics of Calcium Hydroxide in Seawater. *ACS Engineering Au*, *4*(4), 422–431. https://doi.org/10.1021/acsengineeringau.4c00008

Yakaboylu, G. A., Baker, D., Wayda, B., Sabolsky, K., Zondlo, J. W., Shekhawat, D., Wildfile, C., & Sabolsky, E. M. (2019). Microwave-Assisted Pretreatment of Coal Fly Ash for Enrichment and Enhanced Extraction of Rare-Earth Elements. *Energy & Fuels*, *33*(11), 12083–12095. https://doi.org/10.1021/acs.energyfuels.9b02846

d. 1D Risks:

What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Our approach taps into **multiple value streams**–critical metals, reactive hydroxides, and waste remediation–to lower overall project risk. Nonetheless, our research includes targeted efforts to constrain key risks associated with both scientific and economic uncertainties. MRV risk is comparatively lower for well-engineered mineralization systems, compared to many CDR pathways. MRV approaches for OAE are evolving; as they necessarily require global ocean modeling, these risks are beyond the scope of our analysis. Our process focuses on *upstream* considerations for the production of alkaline feedstock, and key uncertainties are scientific and economic.

Key **scientific** risks lie in variable feedstock reactivity, which can affect the efficiency of critical metal and hydroxide recovery. While feedstocks like coal ash and asbestos are highly alkaline, their mineralogy and reactivity can be heterogeneous. The reactivity of coal ash can vary due to initial coal composition, combustion and processing conditions, and storage environments (e.g., fresh resources or impounded). Ash with more glassy mineral qualities requires more energy to extract alkalinity, while more structured minerals are more labile. A central question is whether sufficient reactivity can be achieved with our process, without reagent intensity eroding cost advantages. We will mitigate these risks by testing pre-treatment approaches to broaden the reactivity window and test recovery efficiency across a range of feedstock chemistries, aiming to optimize process conditions for both cost and carbon footprint.

Uncertainty in recovery efficiency leads to **economic** risks, because relative process costs increase as recovery decreases. Equally important, feedstocks vary in concentration of recoverable metals, directly impacting revenue. Our earlier work considering coal ash for CCUS applications found that recoverable metals are a major driver of feedstock viability. Processing costs can reach $1,400 per tonne of ash, so feedstocks with low concentrations of metals or less efficient extraction face profitability challenges. However, some feedstocks have sufficient extractable metals to be profitable: in Figure 5, we show a class C coal fly ash with recovered metal valued at $1,800/t ash processed, leading to optimistic profit metrics.

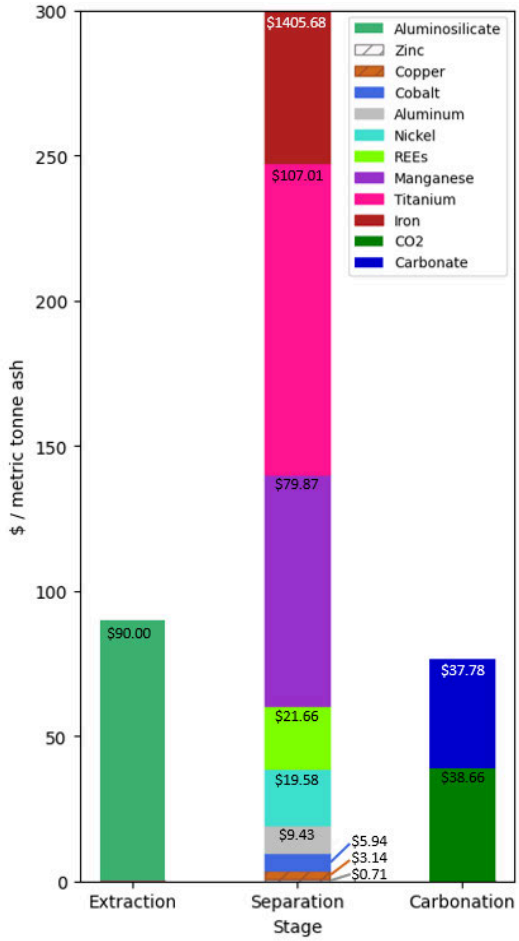


Figure 5. Potential value recovery for the Scherer ash feedstock (class C fly ash) broken down by product.

An additional **economic** risk relates to scaling in a complex industrial ecosystem, where feedstock valuation changes over time. A fraction of coal fly ash has traditionally been beneficially reused (e.g., as cement additives). Figure 6 shows that coal consumption has decreased in recent decades, yet beneficial reuse has remained constant, resulting in decreased availability for alternative uses. The risk of decreased access to fresh feedstock can be partially mitigated by the vast stockpiles of coal ash across the US. An estimated 1.5 billion tons of coal combustion products (CCP) have been landfilled from 1991 to 2016 (DOE, 2022). However, recent reports have noted more than 35 sites across the US with existing or under development projects for harvesting landfilled CCP. The access to 1.5 billion tons of feedstock significantly derisks this process. Asbestos waste is not currently produced, but stockpiles are available at an estimated 2 billion tons (Sandalow et al., 2021).

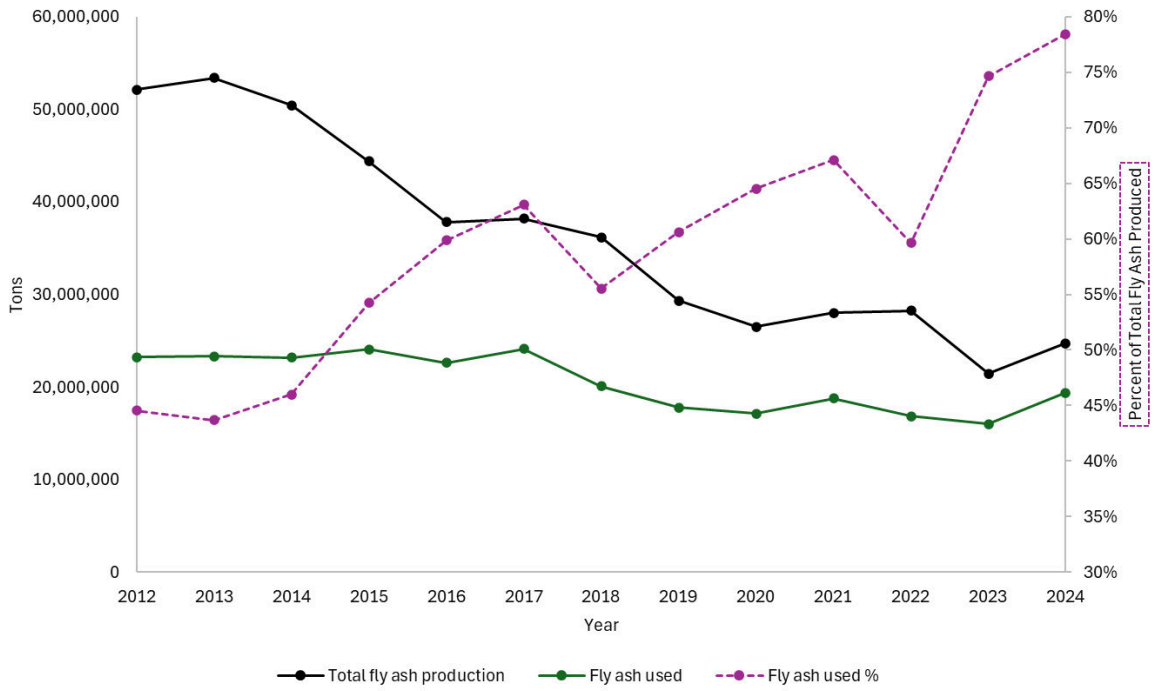


Figure 6. Comparison of total coal fly ash production rates with percentage used.

ACAA. (2025). *Beneficial Use of Coal Combustion Products: An American Recycling Success Story*. https://acaa-usa.org/wp-content/uploads/2022/12/22-ACAA-Brochure_FINAL.pdf

DOE. (2022). *Recovery of Rare Earth Elements and Critical Materials from Coal and Coal Byproducts*. US Department of Energy. https://www.energy.gov/sites/default/files/2022-05/Report%20to%20Congress%20on%20Recovery%20of%20Rare%20Earth%20Elements%20and%20Critical%20Minerals%20from%20Coal%20and%20Coal%20By-Products.pdf

Sandalow, D., Aines, R., Friedmann, J., Kelemen, P., McCormick, C., Power, I., Schmidt, B., & Wilson, S. (2021). *Carbon Mineralization Roadmap*. ICEF Innovation Roadmap Project. https://www.icef.go.jp/wp-content/uploads/2024/02/icef2021_roadmap.pdf

e. 1E Proposal to Frontier:

Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes,*	Bench-scale prototype of the complete workflow of feedstock processing through surficial mineralization, including optimized process conditions

or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)	for a range of feedstocks. Peer-reviewed techno-economic and life-cycle analysis models of co-recovery of critical metals and hydroxides from industrial waste feedstocks.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	This work resolves two key bottlenecks for surficial mineralization: 1) Cost of mineralization is prohibitive for most feedstocks, and we systematically identify what feedstocks, in which locations, are cost-competitive when co-recovery of critical metals is included. 2) Our experimental work provides mechanistic understanding of how carbonation proceeds in mixed CaMg(OH)2 – as opposed to pure hydroxides -- which can have implications for a wide variety of synthetic feedstocks.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	The project will be completed within ~15 months (September 2026 – December 2027), with experimental work in the first 6–8 months followed by integrated modeling and scaling analysis.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$390,620 total grant request, allocated across milestone-based payments tied to experimental validation, pretreatment optimization, and integrated technoeconomic and geospatial analysis.